

Contact-Topology Manifolds in a Combined Finger–Object State Space: A Geometric Representation for Active-Surface In-Hand Manipulation

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Abstract—Controlled sliding of an object along the finger surface is routine in human manipulation and is now physically realizable by active-surface grippers that actuate the finger itself through belts, rollers, or friction modulation. Yet this capability remains *inexpressible* to most in-hand manipulation planners, which parameterize contact solely on the object surface. During sliding, the contact evolves on the *finger*, not on the object, so object-centric representations render such motion invisible, fragment physically continuous motions into disconnected grasp modes, and ignore geometry-dependent centroid drift that leads to collisions with the hand. We argue that this is a representational failure rather than an algorithmic one, and propose planning in a combined finger–object state space $\mathbb{R}^{2n} \times S^3 \times \mathbb{R}^{n_q}$ in which the object centroid is projected into local frames attached to each finger link while orientation is carried relative to the palm. This space is high-dimensional but strongly structured: each contact topology—point-, line-, and plane-on-plane—defines a constraint manifold $\mathcal{M}_\tau$ that collapses $SE(3)$ object motion onto a low-dimensional feasible set. Manipulation primitives then partition naturally into topology-preserving motions that evolve *within* a manifold and topology-switching motions that act as deterministic transitions *between* adjacent manifolds, turning planning into an A^* search over a purely geometric manifold roadmap. Because the abstraction is geometric, it is hardware-agnostic. We validate it on two mechanically distinct active-surface grippers: a variable-friction gripper in hardware (planning time mean < 3 s and planning success 100% across six geometries) and a belt-driven gripper in simulation (100% success versus 50% and 33% for object-centric baselines).

Index Terms—In-hand manipulation, constraint manifolds, multimodal planning, contact topology, active-surface grippers, geometric representations.

I. MOTIVATION: A REPRESENTATION PROBLEM, NOT A SEARCH PROBLEM

Dexterity—repositioning and reorienting an object within the hand—has historically been pursued through anthropomorphic, highly articulated designs whose planning complexity limits deployment [1], [2]. Simpler non-anthropomorphic hands reduce that complexity, but most work on them remains centered on fingertip-dominated rolling contact, leaving *controlled sliding* largely underexplored.

This omission is encoded in the representation itself. Classical in-hand planners—kinematic [17], quasi-static [7], dy-

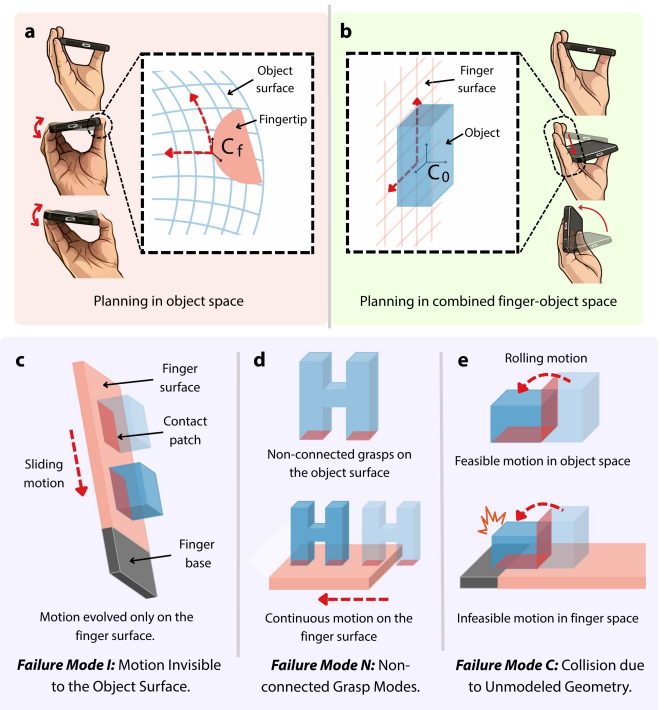


Fig. 1. **Object-centric versus combined finger–object planning.** Conventional in-hand manipulation planners represent contact on the object surface (a). Our formulation instead represents the object relative to the finger surface (b), explicitly capturing translation along the finger surface. (c–e) Failing to jointly model the coupled finger–object state induces recurring failure modes (I, N, C), as detailed in the Introduction.

namic, and recent trajectory-optimization methods [20], [27]—assume rolling-only fingertip–object interaction and therefore trace contact trajectories on the *object* surface. Under this assumption slip is a control failure to be suppressed, rather than a manipulation primitive to be planned with.

Active-surface grippers invert this premise. By actuating the finger surface through controllable friction [13], [15], [21], [24], belts [11], [14], [19], [25], [26], or rollers [23], [28], [29], they make sliding deliberate and repeatable. During slid-

ing/translation along the finger surface, the contact region *on the object* frequently remains fixed while the contact sweeps *along the finger*. The resulting trajectory has no image on the object surface. We identify three recurring failure modes:

- **Mode I — Motion invisible to the object surface.** Pure sliding along the finger surface produces no change in object-surface contact coordinates; an object-centric planner cannot represent, and therefore cannot plan, the motion at all.
- **Mode N — Non-connected grasp modes.** When the planner state is parametrized on the object surface, physically continuous sliding appears artificially disconnected near sharp corners, high-curvature regions, or under heuristic grasp-validity filters. Xie et al. [26] report exactly this: heuristic sampling can reject grasps that are in fact valid, preventing the planner from connecting states the hardware can traverse.
- **Mode C — Collision from unmodeled geometry.** Rotations about object corners induce a *geometry-dependent* translation of the object centroid along the finger. Planners reasoning only about object pose (e.g., [18], which relies on hand compliance) implicitly assume all primitives are globally reachable, and generate object trajectories that collide with the gripper base.

Our position is that these are not failures of search efficiency to be patched with better heuristics or denser sampling. They are artifacts of choosing coordinates on the wrong surface. Combined finger–object coordinates are not merely an alternative parameterization—for active-surface hands they are the geometrically correct one.

II. COMBINED FINGER–OBJECT STATE REPRESENTATION

Consider a hand–object system with n rigid finger links driven by n_q joints. Let F_i be the frame at the base of the i th link, F_O the frame at the object centroid, and F_P the palm frame. For each link we project the object centroid onto the y – z plane of F_i , yielding $P_{F_i} \in \mathbb{R}^2$: the location of the object along the length and width of that finger surface. Collecting these projections with the object orientation gives

$$s = (P_{F_1}, P_{F_2}, \dots, P_{F_n}, q_O, Q_H) \in \mathbb{R}^{2n} \times S^3 \times \mathbb{R}^{n_q}, \quad (1)$$

where q_O is a unit quaternion expressing the object’s orientation in $SO(3)$ relative to F_P , and Q_H is the gripper joint configuration.

The projections P_{F_i} are related through object pose and hand kinematics and are thus not independent. We deliberately retain this redundant parameterization: it makes the object’s position relative to *every* finger link explicit, which is what contact evaluation, finite-surface bounds, and collision checks actually need. Given known hand and object geometry, s uniquely determines the contact locations on the object (expressed in F_O) together with the contact topology τ_i at each finger–object interface. Translational motion is bounded by the finite extent of the finger surface, which directly rejects states

in which the object leaves the grasp or penetrates the gripper base.

For a two-fingered parallel-jaw variable-friction (VF) gripper this reduces to $s = (d_L, d_R, d_Z, q_O, Q_H)$, with d_L, d_R the centroid projections along the left and right fingers and d_Z a shared height coordinate—a form that is convenient for visualizing the constraint manifolds directly.

We assume: no contact breaking (each finger keeps at least one contact); known hand and object geometry, with finger links modeled as convex cuboids and objects as polygonal meshes; and quasi-static dynamics.

III. CONTACT TOPOLOGY MANIFOLDS

The representation above is expressive but high-dimensional. The structure that makes it tractable comes from separating two components of the contact constraint that operate at different levels.

Following the constraint-state formulation of Salisbury, Brock, and Mason [4], [16], the admissible motion of a grasped object is determined by the location, topology, and friction state at each contact. Because active surfaces interact through extended planar finger surfaces rather than fingertip points, we model each finger link as a plane and admit four topologies per interface: point-on-plane, line-on-plane, plane-on-plane, and no contact. For grippers capable of actively modulating contact friction, the effective coefficient of friction can be approximated as *contact without friction* during planning. Conversely, the high-friction state is modeled as a *contact with friction*. In contrast, grippers that apply tangential thrust on the object, such as belt- or roller-operated, maintain high-friction contacts throughout manipulation. Thus, each active contact carries a friction state f_i (with or without friction), giving six distinct contact types. The contact location is already encoded as a coordinate of the state s via the centroid projections P_{F_i} ; therefore, it enters planning as a coordinate on the manifold rather than as part of the manifold-defining constraint.

The key observation is this: **contact topology is purely geometric, while friction is mechanism-dependent.** Topology determines the set of admissible configurations and therefore *defines the manifold*; friction determines which motions are admissible *within* it. Exploiting only the topological component yields a gripper-independent high-level planning abstraction. Writing $\tau = (\tau_1, \dots, \tau_n)$, each topology defines a contact topology manifold

$$\mathcal{M}_\tau = \{s \in S \mid F_\tau(s) = 0\}, \quad (2)$$

where F_τ encodes non-penetration together with the rotational restrictions implied by the topology—coplanarity for plane-on-plane, edge alignment for line-on-plane. This is precisely the constrained-manifold setting studied in multimodal planning [10], [12]: an intractable search in a high-dimensional space becomes a search over discrete modes, each confining motion to a lower-dimensional feasible set.

Manipulation primitives then classify by their action on these manifolds:

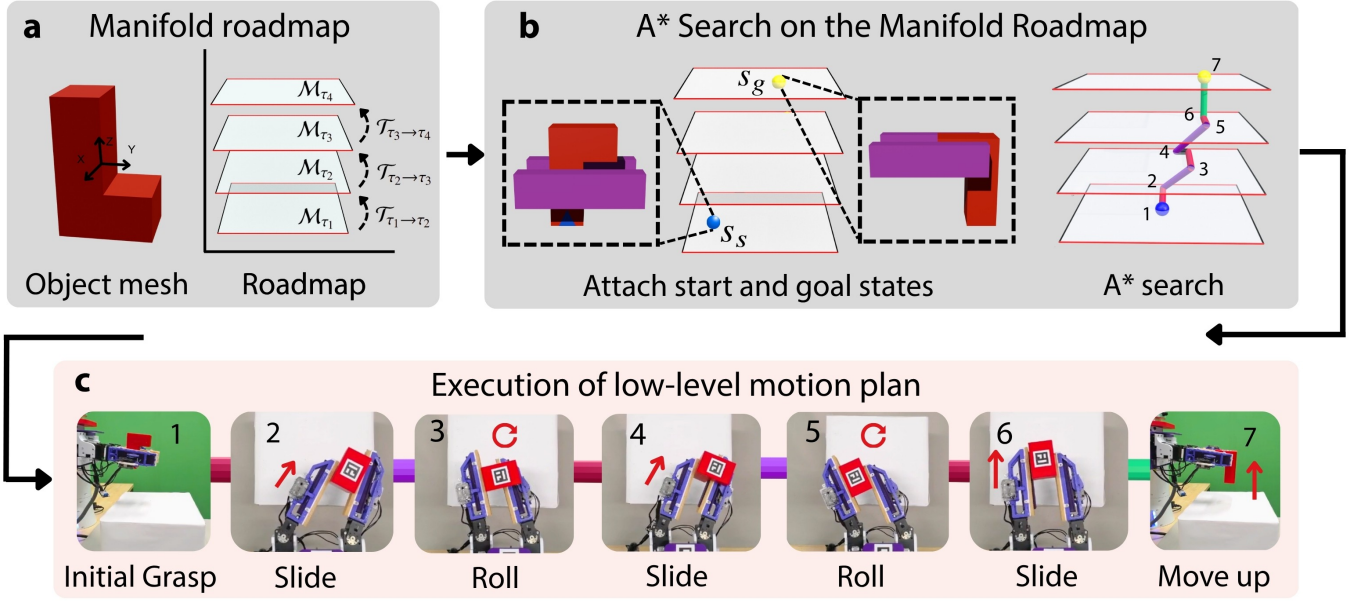


Fig. 2. **Hierarchical planning over contact topology manifolds.** (a) A manifold roadmap is constructed from object geometry alone, connecting feasible contact topologies independently of any actuation mechanism. (b) A* search selects a sequence of manipulation primitives, alternating between topology-preserving motion *within* a manifold $\mathcal{M}_\tau$ and topology-switching transitions $\tau_{\tau \rightarrow \tau'}$ *between* adjacent manifolds. (c) Each primitive is realized through constraint-level projection, recovering joint configurations that satisfy the full contact constraints. An executed multimodal path (nodes 1–7) is shown on the VF gripper.

- **Topology-preserving primitives** $\mathcal{A}_{\text{preserve}}$ evolve the state *within* $\mathcal{M}_\tau$. Contact locations P_{F_i} and joints Q_H change; the topology does not. Examples: asymmetric sliding on the VF gripper, conveying and pitching on the belt-driven gripper.
- **Topology-switching primitives** $\mathcal{A}_{\text{switch}}$ induce discrete transitions, modeled as deterministic maps $\mathcal{T}_a : \mathcal{M}_\tau \rightarrow \mathcal{M}_{\tau'}$. The transition operator encodes the *geometry-dependent displacement* accompanying each topology change—exactly the coupling whose omission produces Mode C. Example: in-hand rotation.

IV. PLANNING: MANIFOLD ROADMAP AND HEURISTIC SEARCH

A. Manifold Roadmap

The first stage builds a roadmap $\mathcal{R} = (\mathcal{T}, \mathcal{E})$ that is *purely geometric*—independent of any primitive or actuation mechanism. Rather than sampling grasps under antipodal validity heuristics [8], [26], we enumerate every candidate assignment of object features (faces, edges, vertices) to fingers and retain those feasible under a single coupling parameter $\theta_{\max} \in [0, \pi]$, the maximum relative tilt between fingers about their closing axes. Edges connect topologies differing in the contacted feature of one finger, with the other fixed. Replacing a grasp heuristic with a geometric enumeration allows us to relax aggressive filtering of possible grasps.

B. A* Search on the Manifold Roadmap

Stage two runs weighted A* over the roadmap. Neighbors are generated by both preserving and switching primitives,

$\mathcal{N} \leftarrow \mathcal{N}_{\text{preserve}} \cup \mathcal{N}_{\text{switch}}$, with candidates validated against finger-surface bounds, collisions, and joint limits. The priority function is $f(n) = g(n) + \lambda_h h(n.s, s_{\text{goal}})$, where the transition cost $c(n, n') = C_{\text{action}} + C_{\text{switch}}$. The C_{switch} adds an explicit penalty for changing primitives, capturing the cost of retracting friction pads or reversing belt direction, which affects execution time and hardware wear.

Because edge costs are temporal while the state is geometric, we evaluate three geometric heuristics defined in the combined space: Manhattan and Euclidean distance over the finger-space projections, $\sum_i \|P_{F_i} - P_{F_i, \text{goal}}\|$ and its ℓ_2 counterpart, and the quaternion distance $d_q = 2 \arccos|\langle q_O, q_{O, \text{goal}} \rangle|$ for orientation guidance. These are not inherently admissible; λ_h maps geometric magnitudes onto the temporal cost scale, trading strict optimality for efficiency [9].

C. Low-level Motion Plan

Given a high-level plan, the object motion is prescribed, and the unknown is Q_H . With $c_i = (\tau_i, f_i)$, where τ_i is the topology at the current state and f_i is the associated friction state of the selected primitive, the $\mathcal{C} = (c_1, \dots, c_n)$ denotes the complete set of constraints that the state should satisfy. We solve $F_{\mathcal{C}}(Q_H) = 0$ by iteratively to recover joint values Q_H using damped pseudoinverse updates [5],

$$Q'_H = Q_H - \alpha J_F^+(Q_H) F_{\mathcal{C}}(Q_H), \quad (3)$$

re-projecting onto joint and palm-width box constraints each iteration. Palm width is itself an optimization variable, letting the solver adapt finger separation to maintain contact across high-curvature regions.

TABLE I
OVERALL PLANNING SUCCESS RATE ON A VARIABLE-FRICTION [21](VF,
HARDWARE) AND A BELT-ORIENTING PHALANGES [26](BOP,
SIMULATION) ACTIVE-SURFACE GRIPPER.

Planner	VF success (%)	BOP success (%)
Object graph (Cruciani et al.)	0.00	33.33
Compliance-based (Morgan et al.)	24.32	50.00
Sampled object contacts (Xie et al.)	0.00	33.33
Contact-topology manifolds (ours)	100.00	100.00

The proposed framework therefore separates high-level topology planning from low-level constraint satisfaction, allowing the same planner to be instantiated across different active-surface grippers by modifying only the manipulation primitive library.

V. RESULTS

All trajectories are executed *open-loop* from known object meshes; fiducial markers are used only for post-hoc error analysis. The intent is to test whether the manifold abstraction *alone* suffices.

Planar ($SE(2)$), VF gripper, hardware. Six geometries (two hexagons, star, rectangle, square, half-curved), three tasks each. Pure 40 mm translation stays on a single manifold for every geometry—motion that is unobservable in object-surface coordinates (Mode I). Coupled translation–rotation tasks produce plans that differ across geometries precisely because corner rotations induce different centroid drift (Mode C). Star and half-curved objects, where object-centric methods fragment continuous sliding (Mode N), are handled without discontinuity. Over five trials per task, translations remain within the ± 5 mm band with 8.73% mean absolute percentage error; discrete face transitions were achieved exactly.

Unique capability. A Rubik’s cube task requires not just a target orientation but a target *position within the grasp*, so that a chosen layer protrudes and becomes accessible. The planner translates the cube toward a pinch-like grasp to expose the layer, slides it back for rotation clearance, reorients, and slides outward again.

$SE(3)$, VF gripper. Across 10 tasks of increasing action count on a hexagonal prism, Dijkstra planning time grows sharply, while all three informed heuristics improve on it; Euclidean and Manhattan distance in finger space bound planning below 7 s across all tasks, with quaternion distance competitive but more variable. On convex, non-convex, asymmetric, and YCB objects, planning times range 0.37–3.78 s with position errors of 0.21–0.63 cm and zero failures on rigid objects for physical experiments. Elevated error on the gelatin and pudding boxes traces to internal mass redistribution during pivoting—a dynamic effect outside the quasi-static model.

Generality: belt-driven (BOP) gripper, simulation. The same formulation, instantiated with a different primitive library (convey, rotate, pitch), yields a different manifold structure. Against baselines adapted to the same hardware, our planner succeeds on all task categories while Cruciani et al. [8] fails

through Mode I, Xie et al. [26] through Modes I and N, and Morgan et al. [18] through Mode C. Overall success: 100% on both platforms, versus 50% and 33% for the strongest baselines, as summarized in Table I.

VI. DISCUSSION AND OPEN QUESTIONS

Three properties of the representation explain these results, and each connects to broader themes in geometric robotics.

Constraints are encoded, not enforced. Primitives that restrict contact topology of active contacts *are* submanifold motions; primitives that switch contact topology *are* manifold transitions. This yields dimensionality reduction and structured connectivity, where object-centric coordinates demand either exhaustive contact-mode enumeration [6] or random grasp sampling [26]—both scaling poorly.

Continuity is restored. Actions that appear discontinuous in object-centric coordinates are continuous paths along the finger surface. The convexity of the finger surface smooths transitions across high-curvature object regions, relaxing strict normal-alignment criteria and naturally connecting grasp regions that heuristic validity tests would separate.

The abstraction is hardware-agnostic. Friction modulation and belt actuation share no mechanism, yet share a manifold structure. The proposed planning framework separates *what state change is required* from *which motor commands produce it*. Emerging mechanisms—soft pneumatic rollers, gecko-inspired adhesives, hybrid friction-belt systems—should be usable without redesigning the planner.

We see several questions worth discussion at the workshop. First, the framework assumes known meshes, and integrating proprioceptive [3] or visuotactile [22] shape estimation means constructing the manifold roadmap under geometric uncertainty—a natural meeting point between manifold methods and tactile sensing. Second, grasp acquisition is currently a separate preceding step; jointly selecting an initial grasp *on a manifold close to the goal* is a backward-planning problem we have not yet solved. Finally, the quasi-static assumption breaks for objects with shifting internal mass, as our own error analysis shows, suggesting a role for grasp mechanics formulations that carry an impression of grasp stability on the same manifold structure.

SUPPLEMENTARY MATERIAL

Video of hardware execution with synchronized manifold visualizations is included with the submission.

REFERENCES

- [1] A. Bicchi. Hands for dexterous manipulation and robust grasping: a difficult road toward simplicity. *IEEE Transactions on Robotics and Automation*, 16(6):652–662, December 2000.
- [2] A. G. Billard. In good hands: A case for improving robotic dexterity. *Science*, 386(6727):eadu2950, December 2024.
- [3] Igor Bodnar and Adam J Spiers. Proprioceptive object shape and size extraction via in-hand-manipulation with a variable friction robot gripper. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 14270–14276. IEEE, 2025.
- [4] David Lawrence Brock. Enhancing the dexterity of a robot hand using controlled slip. In *Proceedings. 1988 IEEE International Conference on Robotics and Automation*, pages 249–251. IEEE, 1988.

- [5] Samuel R Buss and Jin-Su Kim. Selectively damped least squares for inverse kinematics. *Journal of Graphics tools*, 10(3):37–49, 2005.
- [6] Xianyi Cheng, Eric Huang, Yifan Hou, and Matthew T Mason. Contact mode guided motion planning for quasidynamic dexterous manipulation in 3d. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 2730–2736. IEEE, 2022.
- [7] Moëz Cherif and Kamal K Gupta. Planning quasi-static fingertip manipulations for reconfiguring objects. *IEEE Transactions on Robotics and Automation*, 15(5):837–848, 1999.
- [8] Silvia Cruciani, Christian Smith, Danica Kragic, and Kaiyu Hang. Dexterous manipulation graphs. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 2040–2047. IEEE, 2018.
- [9] Eric A Hansen and Rong Zhou. Anytime heuristic search. *Journal of Artificial Intelligence Research*, 28:267–297, 2007.
- [10] Kris Hauser and Victor Ng-Thow-Hing. Randomized multi-modal motion planning for a humanoid robot manipulation task. *The International Journal of Robotics Research*, 30(6):678–698, 2011.
- [11] Yuzuka Isobe, Sunhwi Kang, Takeshi Shimamoto, Yoshinari Matsuyama, Sarthak Pathak, and Kazunori Umeda. Vision-based in-hand manipulation for variously shaped and sized objects by a robotic gripper with active surfaces. *IEEE Access*, 11:127317–127333, 2023.
- [12] Zachary Kingston, Mark Moll, and Lydia E Kavraki. Sampling-based methods for motion planning with constraints. *Annual review of control, robotics, and autonomous systems*, 1(1):159–185, 2018.
- [13] Qiujie Lu, Angus B Clark, Matthew Shen, and Nicolas Rojas. An origami-inspired variable friction surface for increasing the dexterity of robotic grippers. *IEEE Robotics and Automation Letters*, 5(2):2538–2545, 2020.
- [14] Raymond R Ma and Aaron M Dollar. In-hand manipulation primitives for a minimal, underactuated gripper with active surfaces. In *International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, volume 50152, page V05AT07A072. American Society of Mechanical Engineers, 2016.
- [15] Shambhuraj Mane, Anuj Jagetia, Samruddhi Naukudkar, Andrew S Morgan, and Berk Calli. Vibration-induced friction modulation to enable controlled sliding for in-hand manipulation. In *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 142–149. IEEE, 2025.
- [16] Matthew T Mason and J Kenneth Salisbury Jr. *Robot hands and the mechanics of manipulation*. MIT press, 1985.
- [17] David J Montana. The kinematics of contact and grasp. *The International Journal of Robotics Research*, 7(3):17–32, 1988.
- [18] Andrew S Morgan, Kaiyu Hang, Bowen Wen, Kostas Bekris, and Aaron M Dollar. Complex in-hand manipulation via compliance-enabled finger gaiting and multi-modal planning. *IEEE Robotics and Automation Letters*, 7(2):4821–4828, 2022.
- [19] Toshihiro Nishimura and Tetsuyou Watanabe. Single-motor robotic gripper with three functional modes for grasping in confined spaces. *IEEE Robotics and Automation Letters*, 8(11):7408–7415, 2023.
- [20] Tao Pang, HJ Terry Suh, Lujie Yang, and Russ Tedrake. Global planning for contact-rich manipulation via local smoothing of quasi-dynamic contact models. *IEEE Transactions on robotics*, 39(6):4691–4711, 2023.
- [21] Adam J Spiers, Berk Calli, and Aaron M Dollar. Variable-friction finger surfaces to enable within-hand manipulation via gripping and sliding. *IEEE Robotics and Automation Letters*, 3(4):4116–4123, 2018.
- [22] Sudharshan Suresh, Haozhi Qi, Tingfan Wu, Taosha Fan, Luis Pineda, Mike Lambeta, Jitendra Malik, Mrinal Kalakrishnan, Roberto Calandra, Michael Kaess, et al. Neuralfeels with neural fields: Visuotactile perception for in-hand manipulation. *Science Robotics*, 9(96):eadl0628, 2024.
- [23] Kenji Tahara, Keigo Maruta, and Motoji Yamamoto. External sensorless dynamic object manipulation by a dual soft-fingered robotic hand with torsional fingertip motion. In *2010 IEEE international conference on robotics and automation*, pages 4309–4314. IEEE, 2010.
- [24] Clark B Teeple, Buse Aktaş, Michelle C Yuen, Grace R Kim, Robert D Howe, and Robert J Wood. Controlling palm-object interactions via friction for enhanced in-hand manipulation. *IEEE Robotics and Automation Letters*, 7(2):2258–2265, 2022.
- [25] Vinicio Tincani, Manuel G Catalano, Edoardo Farnioli, Manolo Garabini, Giorgio Grioli, Gualtiero Fantoni, and Antonio Bicchi. Velvet fingers: A dexterous gripper with active surfaces. In *2012 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 1257–1263. IEEE, 2012.
- [26] Gregory Xie, Rachel Holladay, Lillian Chin, and Daniela Rus. In-hand manipulation with a simple belted parallel-jaw gripper. *IEEE Robotics and Automation Letters*, 9(2):1334–1341, 2023.
- [27] Fan Yang, Thomas Power, Sergio Aguilera Marinovic, Soshi Iba, Rana Soltani Zarrin, and Dmitry Berenson. Multi-finger manipulation via trajectory optimization with differentiable rolling and geometric constraints. *IEEE Robotics and Automation Letters*, 2025.
- [28] Shenli Yuan, Lin Shao, Connor L Yako, Alex Gruebele, and J Kenneth Salisbury. Design and control of roller grasper v2 for in-hand manipulation. In *2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 9151–9158. IEEE, 2020.
- [29] Shenli Yuan, Shaoxiong Wang, Radhen Patel, Megha Tippur, Connor L Yako, Mark R Cutkosky, Edward Adelson, and Kenneth Salisbury. Tactile-reactive roller grasper. *IEEE Transactions on Robotics*, 2025.